

SLIM

Source Library Manager

Release 2
Version 1.0

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SLIM R2 V1.0 - A package to provide Source Library Management
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Table of Contents

Introduction.....	5
SLIM Basics.....	5
SLIM vs. GIT.....	6
Prerequisites for Installing SLIM R2 V1.0.....	6
Installing/Upgrading SLIM.....	6
Configure SLIM.....	7
Step 1 – Create Archive and Profile.....	7
Step 2 – Create SLIMAUTH.....	7
Step 3 – Create SLIMDDN.....	7
SLIM JCL Templates.....	8
Finishing Up.....	8
Dependencies.....	9
Special Note for CLEANUP.....	9
Using SLIM.....	10
General Usage.....	10
Proc Keyword Definition.....	11
Proc Keywords.....	11
SLIMCL/SLIMCLS Proc.....	12
SLIMCLON Proc.....	13
SLIMCOMP Proc.....	14
SLIMDEP Proc.....	15
SLIMEXTR Proc.....	15
SLIMEXT2 PROC.....	16
SLIMEXTC Proc.....	16
SLIMNURP PROC.....	16
SLIMOTR Proc.....	17
SLIMPAN/SLIMPANS Proc.....	17
SLIMREFR Proc.....	18
SLIMSNAP Proc.....	18
SLIMXREF Proc.....	18
Using SLIMTSO.....	20
Main Menu.....	21
Defaults Menu.....	22
Add Notes Panel 1.....	23
Add Notes Panel 2.....	24
Dependencies Panel.....	25
Common Panel.....	28
Batch Menu.....	29
Batch Panels.....	30
Creating a SLIM Group.....	32

SLIM Release 2 Version 1.0
February 17, 2026

Initialize The Group.....	32
ITEMS That Require Special Handling.....	34
Customizing SLIM.....	35
SLIMDDN Table.....	35
SLIMAUTH Table.....	35
Appendix A – The SLIMDDN Table.....	36
Appendix B – The SLIMAUTH Table.....	37

Table of Figures

Figure 1: Sample Group/Subgroup Report.....	19
Figure 2: Sample Subprojec/Group Report.....	19

Introduction

SLIM is an easy to use tool for storing projects. It is an “add on” to the ARCHIVER package to simplify the management of software items and for distribution of software. ARCHIVER is included in the TK5 distribution. ARCHIVER can be found on CBT File 147 at CBTTAPE.ORG.

SLIM (**S**ource **L**ibrary **M**anager) was originally designed as a means of storing production source code and documenting relationships between projects. I view it as a “project dictionary” similar to the way relational databases have a “data dictionary”.

A project consists of one or more modules. A module is a best defined a unit of source code. For example, an assembler macro, a COBOL copy book or an entire source program are each considered modules in SLIM. In addition, source listings and copies of load modules may be added into a SLIM archive as modules.

With this SLIM release, a COMMON type of project was added. Common projects are used for projects that are used in many places. Some examples of common code are macros used in many projects, interfaces to VSAMIO for OS/360 compilers, COBOL or PL/1 record descriptions used in several projects, etc.

ARCHIVER uses VSAM to store the items in an encrypted and compressed format. It is not easy to change items except through ARCHIVER or SLIM software.

With the last release of SLIM, an “online” component was added called SLIMTSO. This component provides interactive interfaces to SLIM. SLIMTSO can perform many SLIM functions either online or by submitting a batch job.

Beside the renumbering, this SLIM Release adds, some enhancements were added and some bugs fixed. Changes include:

- Common Module supported
- JCL template jobname generation
- Bug fixes

SLIM Basics

SLIM organizes modules using five factors – ITEM, GROUP, SUBGROUP, TYPE and VERSION. In SLIM V1.x.x, this document refereed to these fields as MODULE, PROJECT, DESCRIPTION, TYPE and VERSION. Through out this document, I will use the ARCHIVER names rather than the SLIM names.

SLIM provides many routine processes to provide simple but effective management of software items. Items are compressed and encrypted to provide a basic level of protection. SLIM has the ability to

store different types of data for a project. The source code, which includes macro, copy items, compiler generated source listings as well as load modules.

There are ways other than SLIM to access an Archive. The SLIM software does not provide full maintenance support for a VSAM archive. IDCAMS and ARCHIVER software provide most of the maintenance of the physical copy of the VSAM archive. SLIM provides the maintenance of the logical organization of the archive.

RPF (Rob's Programming Facility) provides an excellent TSO interface to VSAM archives. RPF (version V1R9M9 or latter) provides the ability to perform a number of functions on an archive in an ISPF like manner. RPF is preinstalled in Tk5 update 3 or latter. TK5 can be downloaded from <https://www.prince-webdesign.nl/rpf>.

SLIM vs. GIT

If you are familiar with a version control system for PCs called GIT. GIT provides a checkout/checkin function for complete projects. A complete project consists of all code and files to provide a private copy of all pieces of the project. GIT provides tracking of changes so when a project is checked in, it checks if any subprojects are affected. SLIM provides *some* functions similar to GIT. SLIM provides a project based staging/commit facility similar to GIT. SLIM does not provide sophisticated check in/out facilities like GIT does.

Prerequisites for Installing SLIM R2 V1.0

- Downloaded and installed TK5 update 4. Note TK5 update 4 has a previous version installed.

or

- Any Other installation of Hercules and MVS 3.8j
- ARCHIVER 6.2.0 (can be found on CBT File 147 at CBTTAPE.ORG)
- MAP3270 V3.2.3
- ARCHEXTN V1.x.x or later
- unzip program

Installing/Upgrading SLIM

Use of the \$README in the zip you downloaded is the easiest (and recommended) way to install or upgrade SLIM. The RESTTK5 in the zip is the JCL will apply the release. Upon completion of the install, continue with the next section – configure SLIM.

Configure SLIM

Before you can use SLIM, it must be configured. Configuration consists of creating a SLIM ARCHIVE, a SLIM profile, a SLIMDDN module and a SLIMAUTH module.

An archive is a VSAM Keyed Sequential Data Set (KSDS). An archive must be created using the IDCAMS utility.

Step 1 – Create Archive and Profile

If you have an existing SLIM Archive set up, skip to Step 2.

The JCL for doing this can be found in SLIM.CNTL(MAKEARCH). This is the Archiver authors recommended options for an optimal archive. Be sure and make any suggested changes in the JCL. This JCL will create both an ARCHIVE and a PROFILE. If an ARCHIVE already exists, follow the instructions in the JCL to skip re-creating the ARCHIVE.

Step 2 – Create SLIMAUTH

SLIM dynamically allocates ARCHIVER files using a builtin security system. All executions of SLIM require a HLQ (high level qualifier) and a USER KEY. HLQ is usually a TSOID or catalog alias (i.e. HERC01 or SYS2). USER KEY is a unique name assigned to each user authorized to access SLIM archives. It can be any combination of up to 8 characters. There is a module called SLIMAUTH that verifies the HLQ and USER KEY to determine the dataset name of the ARCHIVE to be accessed.

The SLIMAUTH as delivered has HLQ of HERC01 defined and a USER KEY of DEFAULT defined to access archive 'HERC01.SLIM.ARCHIVE'.

If you have modified SLIMAUTH, you may skip to Step 3. However, I do recommend updating the new SLIMAUTH to match your revisions.

Step 3 – Create SLIMDDN

SLIM organizes items by description and type. This table defines the dataset name where an item will be located. SLIMDDN is used to enforce the per-defined organization. For example, a COBOL source module will be described as SOURCE with type COB and will be located in dataset name 'hlq.group.ddname(item)'.

If you have modified SLIMDDN, skip to Step 4. However, I do recommend updating the new SLIMDDN to match your revisions.

The JCL for doing this can be found in SLIM.CNTL(MAKEDDN). This is a default table. This will cover many situations that may be encountered.

SLIM JCL Templates

Template PDS support has been removed. SLIM templates are stored in the .PROFILE profile. The user profiles are maintained using SLIMTSO (see page 20).

Finishing Up

At this point, all the procs in 'SLIM.PROCLIB' can be copied to your system proclib ('SYS2.PROCLIB'). All the executables in SLIM.LOADLIB can be copied to 'SYS2.LINKLIB' or 'SYS2.CMDLIB'. JCL to do this is can be found in SLIM.CNTL(FINISH).

Dependencies

SLIM is dependent on other projects which are freely available. The full source code and executable for each is included

JOBABEND	This is one of my projects. Abends the job.
SETRC	This is one of my projects. Sets the step return code based on EXEC statement parm. Useful for running or skipping subsequent steps in the job.
MAKESTMT	This is one of my projects. Creates a sequential file from data passed in the parms.
MYGENER	This is one of my projects. Like IEBGENER, it can copy and changes record formats $F \rightarrow V$, $V \rightarrow F$. Records are truncated or padded as appropriate.
CLEANUP	Obtained from CBT183 at CBTTAPE.ORG. Deletes cataloged dataset that will be created in the job. This avoids the dreaded "NOT CATLG 2".
VSAMIO	Obtained from JAYMOSELEY.COM SYSCPK dasd volume. Provides access to VSAM files for COBOL and PL/I program.
SCOMPARE	This source code compare program was taken from the Stanford Pascal Compiler. Obtained from JAYMOSELEY.COM SYSCPK dasd volume.

I have included the source code and executables as part of the distribution. In addition, ARCHIVER 6.1.8 and MAP3270 V3.2.2 (or higher) are required to be installed. They are installed in TK5 update 4.

Special Note for CLEANUP

CLEANUP accesses MVS internals and will requires an additional library (SYS1.AMODGEN) to assemble clean. If you choose to assemble it, make sure it is not executed from a PDS that CLEANUP will delete.

Using SLIM

General Usage

If you want to fast track to SLIMTSO, go to USING SLIMTSO on page 20.

There are several JCL procedures to implement SLIM. Listed in the table below are the procs and corresponding SLIMTSO options. SLIMTSO builds JCL and submits the JCL for batch processing.

SLIMTSO	SLIM Proc	Action
B.3	SLIMCL	Compile and Link without storing results in an archive.
B.4	SLIMCLS	Compile, Link and Store in an archive.
	SLIMCOMP	Proc to Compare two source items.
2	SLIMDEP	Define a dependencies between a project and sub project
B.6	SLIMEXTC SLIMEXTR SLIMEXT2	Proc SLIMEXTC extracts common projects. Proc SLIMEXTR extracts all the items associated with a given project and generates JCL to compile and link all items using SLIMEXT2.
B.5	SLIMOTR	add items other than source code (i.e. copy items, jcl)
B.1	SLIMPAN	Compile a Panel
B.2	SLIMPAN	Compile a Panel and store a MAP3270 panel in an archive
	SLIMREFR	Refresh a given set of items for a specific description
	SLIMRST	Reset version numbers.
	SLIMSNAP	Create an “image” of a project.
	SLIMXREF	Print cross reference reports.

SLIM is invoked as a batch program using procs. All information is declared using keyword on the procs.

Each generated JCL job starts with an execution of CLEANUP. Cleanup examines the JCL and will delete any cataloged sequential or PDS that had a DISP of NEW before the balance of the job executes. GDG and VSAM datasets are NOT processed by CLEANUP.

For any of the SLIM procs, a job card is required. If at installation, you choose to NOT copy the load modules to SYS2.LINKLIB, a JOBLIB is required.

```
//jobname JOB
```

```
//JOBLIB DD DSN=SLIM.LOADLIB,DISP=SHR
```

Since SLIM is managed by SLIM, I will use it for examples through out this document.

Proc Keyword Definition

All of the procs have keywords. The same keywords used in each proc have the same meaning. Some procs have unique keywords and will be documented there.

Proc Keywords

CMD is the command – must be ADD or UPDATE

COPYx provides for user supplied COPY or INCLUDE PDS for the compile.

COPYBUFF is set up to default for TK5 3390 volumes. This allows the concatenation of partition datasets (PDS) of different BLKSIZEs.

DESC is the SUBGRP of the item. See Appendix A on page 36 for a list of valid description or types.

HLQ is the high level qualifier of the SLIM archive to be processed. This is usually a TSO ID or an alias.

IN is the DSN including the item to be compiled.

LANG specifies the language of the module. Valid values are ASM, COB, FOR, PAS and PLI.

LOADLIB is the DSN of the PDS where the LOADMOD will be placed.

LOADMOD is the name of the item to be linked into LOADLIB.

LVL is only used for FORTRAN programs to indicate if the G or H Fortran compiler should be used.

MODULE is the name of the item to be compiled.

NCAL specified to the module should be linked with the NCAL or CALL option.

NCALIBx sets up for user and system runtime libraries used to resolve link edits.

PLIMAC specifies if the MACRO or NOMACRO option should be used for compiling a PL/I F program.

PLISORT is used only for PL/1 programs that invoke the sort/merge program. Link edits for these program result in return code 4 due to the way the sort interface was link edited by IBM. For this case, the RC=4 is considered normal.

PROJECT is the name of the group.

USERID is the SLIM User Key. It is the NOT the TSO used id.

VOL is the DASD volume to allocate datasets on.

SLIMCL/SLIMCLS Proc

The SLIMCL proc will compile and link a program. It is similar to SLIMCLS except the source code, listing and load module are not stored in an archive.

The SLIMCLS proc will compile and link a program. If all steps were successful, the source code, the load module and compiler listing will be stored in the SLIM archive.

```
SYS2.PROCLIB (SLIMCL) -----
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMCL  PROC  SOUT='*',      DEFAULT SYSOUT CLASS                                V2.0.0
//
//          CMD=,
//          HLQ=,
//          USERKEY=,
//          GROUP=,
//          ITEM=,              MODULE TO BE COMPILED
//          DESC=,
//          IN=NULLFILE,        PDS WITH MEMBER TO COMPILE
//          COPY1='SYS1.MACLIB',
//          COPY2='SYS1.MACLIB',
//          COPY3='SYS1.MACLIB',
//          COPY4='SYS1.MACLIB',
//*          COPYBUF=19040,      DEFAULT FOR 3350 DASD
//          COPYBUF=27920,      DEFAULT FOR 3390 DASD TK5
//          LOADLIB=NULLFILE,    PDS TO GET LKED OUTPUT
//          LOADMOD=NULLFILE,    MEM IN PDS TO GET LKED OUTPUT
//          NCAL=,
//          NCALIB1='SYS1.PL1LIB',
//          NCALIB2='SYS1.PL1LIB',
//          COBLIB='SYS1.COBLIB',
//          PLILIB='SYS1.PL1LIB',
//          FORTLIB='SYS1.FORTLIB',
//          PLIMAC='',
//          PLISORT='0,EQ',      FOR SORT PROBLEM USE '5,GT'.
//          LANG=,              3 CHARACTER CODE FOR LANG TYPE
//          LVL=                LANG LEVEL IF MORE THAN 1
//*****
```

SLIMCLON Proc

The SLIMCLON proc is used to make a copy of a group in the Archive.

```
SYS2.PROCLIB(SLIMCLON) -----
                                     BNDS 001,072 SCRL CSR
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMCLON PROC SOUT='*',                                     V2.0.0
//          HLQ=,                HIGH LEVEL QUAL. FOR SLIM ARCHIVE
//          USERKEY=,           SLIM USERID
//          OLDGRP=,
//          NEWGRP=
//*****
```

SLIMCOMP Proc

The SLIMCOMP proc is used to compare to two items. Input to the compare can be any combination of PDS items and/or ARCHIVE items.

```
SYS2.PROCLIB(SLIMCOMP) 01.00-----
                                           BNDS 001,072 SCRL CSR
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMCOMP PROC SOUT='*',                                           V2.0.0
//*****
//*
//* SLIMCOMP - COMPARE SOURCE CODE ITEMS
//* THIS PROC WILL COMPARE SOURCE CODE IN PDS AND/OR
//* ARCHIVER FILES.
//*
//* THIS PROC COMPARES TWO SOURCE FILES. IT PRIMARILY COMPARES A
//* NEW VERSION AND OLD VERSION OF A PROGRAM'S SOURCE CODE. THE
//* INPUT CAN BE FROM A FILE/PDS OR AN ARCHIVER MEMBER.
//*
//*
//*****
//      OLDDSN='*',      |
//      NEWDSN='*',      |
//      OLDARCH='*',      | ONE OLD AND ONE NEW MUST BE CHOSEN
//      NEWARCH='*',      |
//      ITEM='*',         |
//      GROUP='*',        |
//      TYPE='*',         | ALL REQUIRED FOR ARCHIVER MEMBERS
//      OLDLVL=,          |
//      NEWLVL=,          |
//      LANG='*',         | REQUIRED ALWAYS
//      DIFF='*',         | REQUIRED - NUMBER OF COMPARE LINES.
//      PLIS=             | USE ONLY FOR PL1 PROGRAMS WITH SORTS
//*                       | CODE PLIS=S
//*****
```

OLDDSN/NEWDSN specify the PDS(item) to be compared.

OLDARCH/NEWARCH give the specific ARCHIVE name. If either or both are is used, the MEMBER, PROJECT, TYPE and LANG must be given.

DIFF is used by the compare program to determine number of lines to compare to determine difference. 6 is the recommend value.

The process of compare is fairly straight foreword. However, there are many possible combinations of old and new that can be compared.

See the “SLIM.SAMPLES(COMP)” to all the documented source compares that are possible.

SLIMDEP Proc

The SLIMDEP is used to define dependencies between projects. A dependency exists when module that requires another module for successful use. An example is a main program calls a subroutine. That makes the main program dependent upon the subroutine.

```
SYS2.PROCLIB (SLIMDEP) -----  
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--  
***** Top of data *****  
//SLIMDEP PROC SOUT='*', V2.0.0  
//          HLQ=,          HIGH LEVEL QUAL. FOR SLIM ARCHIVE  
//          USERKEY=,      SLIM USER KEY  
//          GROUP=,  
//          SUBGRP=  
//*****
```

In SLIM, as noted above, a project is a collection of items such as source code, copy/include items, macros, etc that are required for the project to be implemented and compiled if necessary. A full explanation of dependencies is discussed under the SLIMEXTR proc.

SLIMEXTR Proc

The SLIMEXTR proc generates JCL to extract all the current versions of items associated with a project as well as dependencies to compile and link a project. This process generates a dataset (userid.project.EXTRACT.CNTL) with JCL statements to extract the items from an archive. No actual extraction takes place in this step. Also, if appropriate, SLIM will generate a dataset (userid.project.COMPILE.CNTL) of JCL with a series of jobs to compile and link ALL of the items in the project.

```
SYS2.PROCLIB (SLIMEXTR) -----  
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--  
***** Top of data *****  
//SLIMEXTR PROC SOUT='*', V2.0.0  
//          HLQ=,  
//          USERKEY=,  
//          GROUP=,  
//          VOL=  
//*****
```

The extract job must be manually submitted. Upon completion, compile job should be submitted. Each job in the compile JCL has the same jobname. This is to ensure the jobs run in sequence of dependency. What this means is subroutines are compiled and linked before any other program calls them. The sequence is determined using the “++DEPEND” items.

For example, Program A calls Sub B. Sub B will be processed first so it is available when Program A is compiled and linked.

SLIMEXT2 PROC

The SLIMEXT2 proc is used to execute the JCL generated by the SLIMEXTR process. The JCL for executing this proc is generated by SLIM and no further explanation will be made.

There are items in the SLIM distribution that require special handling for compiling. See page 34

SLIMEXTC Proc

The SLIMEXTC proc is used to executes the JCL to extract a current version of a COMMON item. This process generates a dataset (userid.project.EXTRACT.CNTL) with JCL statements to extract the items from an archive.

The biggest difference between SLIMEXT2 and SLIMEXTC is SLIMEX2 creates new datasets while SLIMEXTC stores members in existing datasets.

SLIMNURP PROC

```
SYS2.PROCLIB (SLIMNUPR) -----
                                     BNDS 001,072 SCRL CSR
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMNUPR  PROC SOUT='*',                                     V2.0.0
//      GROUP=,
//      USERKEY=,
//      VOL=
//*****
```

The SLIMNUPR proc is used in the create or recreate a project. All possible empty datasets will be created. The DSN will be in the form userid.project.type on volume VOL

SLIMOTR Proc

```
SYS2.PROCLIB(SLIMOTR)-----
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMOTR PROC SOUT='*',                                V2.0.0
//
//          CMD=,
//          IN=NULLFILE,
//          HLQ=,                                HIGH LEVEL QUAL. FOR SLIM ARCHIVE
//          USERKEY=,
//          ITEM=,                                MEMBER TO BE PROCESSED
//          GROUP=,
//          SUBGRP=,
//          TYPE=,
//          LKED=
//*
//*****
```

The SLIMOTR proc is used to process items into an archive that do not need to be compiled such as JCL, PROCS, copy items or any other “card” image.

SLIMPAN/SLIMPANS Proc

The SLIMPAN proc is used to compile a MAP3270 panel **without storing** results into an archive. The SLIMPANS proc is used to compile a MAP3270 panel **storing** results into an archive.

```
SYS2.PROCLIB(SLIMPAN)-----
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMPAN  PROC SOUT='*',    DEFAULT SYSOUT CLASS                                v2.0.0
//
//          CMD=,
//          M=,                FOR MODEL 5 M=5
//          HLQ=,
//          USERKEY=,
//          PANEL=,
//          GROUP=,
//          IN=,                PDS WITH MEMBER TO COMPILE
//          COPY1='MAP3270.MACLIB',
//          COPY2='MAP3270.MACLIB',
//*          COPYBUF=19040,        DEFAULT FOR 3350 DASD
//          COPYBUF=27920,        DEFAULT FOR 3390 DASD TK5
//          LOADLIB=NULLFILE     PDS TO GET LKED OUTPUT
//*****
```

SLIMREFR Proc

```
SYS2.PROCLIB(SLIMREFR) 01.00-----
                                           BNDS 001,072 SCRL CSR
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMREFR PROC SOUT='*',                               V2.0.0
//              ARCH=,                                ARCHIVE
//              LIB=,                                PDS TO REFRESH
//              SPACE='(TRK,(10,10,10))',
//              GROUP=,
//              SUBGRP=,
//              TYPE=
//*****
```

SLIMSNAP Proc

```
SYS2.PROCLIB(SLIMSNAP) 01.01-----
                                           BNDS 001,072 SCRL CSR
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMSNAP PROC SOUT='*',                               V2.0.0
//              HLQ=,
//              USERKEY=,
//              GROUP=,
//              VOL=,
//              LVL=
```

The SLIMSNAP proc produces a snap shot of a project. A snap shot is all the JCL to extract and compile a project in its current state. It works like SLIMEXTR except the generated JCL is stored in an archive. This enables the state of a project be captured. Items in an archive are identified by MEMBER, PROJECT, SUBPROJECT, TYPE and version. Each time an item is updated, the version becomes 1 greater. This allows history of changes to be kept.

SLIMXREF Proc

The SLIMXREF proc print two reports – Group/Subproject which lists all the items in a project. The other Subproject/Group which lists all the projects where the project is reference.

```
SYS2.PROCLIB(SLIMXREF) 01.00-----
                                           BNDS 001,072 SCRL CSR
-----+-----1-----+-----2-----+-----3-----+-----4-----+-----5-----+-----6-----+-----7--
***** Top of data *****
//SLIMXREF PROC SOUT='*',                               V2.0.0
//              ARCH=                                SLIM ARCHIVE TO XREF
//*
```

SLIM Release 2 Version 1.0
February 17, 2026

SLIMXREF V1.0.0		PROJECT/SUB PROJECT XREF				FEBRUARY 16, 2024				PAGE 00001
PROJECT	SUBPROJECT	MODULE	DESC	TYPE	VER	MODULE	DESC	TYPE	VER	
ARCHDATA		ARCHDIR	INCLLIB	PLI	1	ARCHREC	COPYLIB	COB	1	
		ARCHREC	INCLLIB	PLI	1					
ARCHEXTN		ARCHCOMP	LOADLIB	EXEC	1	ARCHCOMP	PROCLIB	JCL	1	
		ARCHCOMP	SOURCE	PLI	1	ARCHCOMP	SYSOUT	LISTING	1	
		ARCHDIR	LOADLIB	EXEC	1	ARCHDIR	PROCLIB	JCL	1	
		ARCHDIR	SOURCE	PLI	1	ARCHDIR	SYSOUT	LISTING	1	
		ARCHUTIL	LOADLIB	EXEC	1	ARCHUTIL	PROCLIB	JCL	1	
		ARCHUTIL	SOURCE	PLI	1	ARCHUTIL	SYSOUT	LISTING	1	
		ARCHUTIU	PROCLIB	JCL	1	CLEANUP	INSTALL	CNTL	1	
		COMPILE	INSTALL	CNTL	1	EXTNTRAN	INSTALL	CNTL	1	
		RESTOTHR	INSTALL	CNTL	1	RESTTKS	INSTALL	CNTL	1	
ARCHEXTN	ARCHDATA	ARCHDIR	INCLLIB	PLI	1	ARCHREC	COPYLIB	COB	1	
		ARCHREC	INCLLIB	PLI	1					
ARCHEXTN	DYNALOAD	DYNALDA	LOADLIB	NCAL	1	DYNALDA	SUBPGM	ASM	1	
		DYNALDA	SYSOUT	LISTING	1	DYNALDP	LOADLIB	NCAL	1	
		DYNALDP	SUBPGM	ASM	1	DYNALDP	SUBPGM	ASM	2	
		DYNALDP	SYSOUT	LISTING	1	DYNALDP	SYSOUT	LISTING	2	
		DYNALOAD	LOADLIB	NCAL	1	DYNALOAD	SUBPGM	ASM	1	
		DYNALOAD	SYSOUT	LISTING	1					
ARCHEXTN	PLIEXTEN	SDEBUG	INCLLIB	PLI	1	FIX2STR	INCLLIB	PLI	1	
		FORMDAT	INCLLIB	PLI	1	RNDGEN	INCLLIB	PLI	1	
		SELECT	INCLLIB	PLI	1	TRIM	INCLLIB	PLI	1	
ARCHEXTN	VSAMIO	VSAMIO	COPYLIB	COB	1	VSAMIO	INCLLIB	PLI	1	
		VSAMIO	LOADLIB	NCAL	1	VSAMIO	SUBPGM	ASM	1	
		VSAMIO	SYSOUT	LISTING	1	VSAMIOFB	COPYLIB	COB	1	
		VSAMIOFB	INCLLIB	PLI	1	VSAMIOFB	LOADLIB	NCAL	1	
		VSAMIOFB	SUBPGM	ASM	1	VSAMIOFB	SYSOUT	LISTING	1	
		VSAMIOFB	INCLLIB	PLI	1					

Figure 1: Sample Group/Subgroup Report

SLIMXREF V1.0.0		SUB PROJECT XREF				FEBRUARY 16, 2024				PAGE 00001
SUBPROJECT	PROJECT	MODULE	DESC	TYPE	VER	MODULE	DESC	TYPE	VER	
ARCHDATA	ARCHEXTN	ARCHDIR	INCLLIB	PLI	1	ARCHREC	COPYLIB	COB	1	
		ARCHREC	INCLLIB	PLI	1					
ARCHDATA	ARCHUTIL	ARCHDIR	INCLLIB	PLI	1	ARCHREC	COPYLIB	COB	1	
		ARCHREC	INCLLIB	PLI	1					
ARCHDATA	SLIM	ARCHDIR	INCLLIB	PLI	1	ARCHREC	COPYLIB	COB	1	
		ARCHREC	INCLLIB	PLI	1					
BASALO	BASICMON	BASALO	LOADLIB	NCAL	1	BASALO	SUBPGM	ASM	1	
		BASALO	SUBPGM	ASM	1	BASALO	SYSOUT	LISTING	1	
		BASALOP	LOADLIB	NCAL	1					
BASALO	BASIC1UP	BASALO	LOADLIB	NCAL	1	BASALO	SUBPGM	ASM	1	
		BASALO	SUBPGM	ASM	2	BASALO	SYSOUT	LISTING	1	
		BASALOP	LOADLIB	NCAL	1					
BASALO	BASICMON	BASALO	LOADLIB	NCAL	1	BASALO	SUBPGM	ASM	1	
		BASALO	SUBPGM	ASM	2	BASALO	SYSOUT	LISTING	1	
		BASALOP	LOADLIB	NCAL	1					

Figure 2: Sample Subprojec/Group Report

Using SLIMTSO

SLIMTSO is a menu driven component of SLIM. It operates in partnership with SLIM but shares a few items with SLIM such as the SLIMAUT/SLIMAUTH and SLIMDDN modules.

On line 2, each panel has an option field and a message field. On the bottom line, there is a line stating what PFKs work. Unless noted, the option X is equivalent to PFK3. Undefined options and PFKs are treated as ENTER key.

Some panels implement a “point and click” scheme. When it is active, an instruction will be displayed telling you place the cursor on an non-blank character which is part of an option. Pressing ENTER will result in that option being populated in the fields.

For each TSO session logon, the SYSPRINT file must be assigned. This can be done in your logon clist or via the allocate command.

```
READY  
      alloc f(sysprint) da(*)
```

For each execution of SLIMTSO, it can be started in one of three ways. Option one is used if SLIM is installed in your system (like in SYS2.CMDLIB).

```
READY  
      slimtso
```

Option 2 to enter the call command to what PDS SLIMTSO is installed in

```
READY  
      call 'herc01.slim.loadlib(slimtso)'
```

Option 3 is to create a clist in you logon with the call statement. Assuming a clist called SLIMTSO was create, simply enter:

```
READY  
      slimtso
```

If all go well, you will see a “splash” line. Press ENTER to continue to the sign in screen. The USER KEY was assigned during the SLIM Configuration (see page 7).

Note the following are samples similar to the actual panels. Remember, in this version only model 4 (43 x 80) screens are supported.

SLIM Release 2 Version 1.0
February 17, 2026

Main Menu

----- SLIMTSO MAIN MENU -----

Option ==> _

Connected To SLIM Archive
HERCEL.SLIM.ARCHIVE

- | | | |
|---|-----------|--|
| 0 | DEFAULTS | Alter / Display / Create Session Profile |
| 1 | ADD NOTES | Add Notes To SLIM items(s) |
| 2 | DEPENDS | View / Add Project Dependency |
| 3 | COMMON | Declare a SLIM group as COMMON group |
| B | BATCH | Submit JCL To Perform SLIM Batch Functions |

CLEAR=EXIT PFK03=EXIT

Defaults Menu

```
----- SLIMTSO DEFAULTS -----
Option  ==> _

      DEFAULTS FOR USERID HERCEL  ARCHIVE HERCEL.SLIM.ARCHIVE

      Group                               SLIMTSO

      Sub Group                           SOURCE

      Type                                PLI

      VOL SER                             EVOL01

      JOB Cards For SLIMTSO Jobs.
      //TESTSLT JOB CLASS=B,MSGCLASS=A,MSGLEVEL=(1,1),
      //      USER=HERCEL,PASSWORD=CUL8TR      ,TYPRUN=SCAN
      //JOB LIB DD DSN=HERCEL.SLIM.LOADLIB,DISP=SHR
      /***
      /***

      JOB Cards For SLIM Jobs.
      //SLIM JOB CLASS=B,MSGCLASS=A,MSGLEVEL=(1,1),
      //      USER=HERCEL,PASSWORD=CUL8TR
      //JOB LIB DD DSN=HERCEL.SLIM.LOADLIB,DISP=SHR
      /**
      /**

      CLEAR=EXIT      PFK03=RETURN NO SAVE      PFK12=SAVE
```

None of the fields are required. They serve as optional default values in other SLIMTSO panels.

Entering the Option 'A', the profile is cleared and you can enter a USER KEY. The default profile is delivered with USER KEYS DEFAULT and HERC01.

Once a profile for a USER KEY is defined, it can only be changed when that USER KEY is signed in.

Add Notes Panel 1

```
----- SLIMTSO ADD NOTES -----  
Option  ==> _  
  
                Connected To SLIM Archive  
                HERCEL.SLIM.ARCHIVE  
  
Item...          REQUIRED  
  
Group...         REQUIRED  
  
Sub Group.....  
  
Type.....
```

CLEAR=EXIT PFK03=RETURN NO SAVE PFK12=SAVE RETURN

This panel is used to locate an item in the Archive to add notes to.

This is a “point and click” panel. A list of valid Desc and Type will be displayed on the lower half of the screen. Simply place the cursor on a non-blank of your choice and press enter. That Desc and Type will be copied into the Desc and Type fields.

If all the criteria are satisfied and the item exists in the Archive, the Add Notes Panel 2 will be displayed.

SLIM Release 2 Version 1.0
February 17, 2026

Add Notes Panel 2

```
----- SLIMTSO ADD NOTES -----  
Option ==> _      NOTES FOUND FOR VERSION 6  
  
      CONNECTED TO SLIM ARCHIVE  
      HERCEL.SLIM.ARCHIVE  
  
-SLIMTSO  -SLIMTSO  -SOURCE  -PLI      -      6  PS  FB  
  
250219 HERCEL SLIMTSO,SLIMTSO,SOURCE,PLI
```

CLEAR=EXIT PFK03=RETURN NO SAVE PFK12=SAVE RETURN

The lower part of this panel allows you to enter and notes you wish to add to the item. Please note that notes are cumulative for an item and do not bump up the version number.

Dependencies Panel

----- SLIMTSO DEPENDENCIES -----
Option ==> _ 52 PROJECTS FOUND

CONNECTED TO SLIM ARCHIVE
HERCEL.SLIM.ARCHIVE

SELECT A PROJECT FROM THE LIST BELOW
PLACE CURSOR ON PROJECT NAME AND PRESS ENTER

\$DUMPR	BASIC1UP	DYNWORK	MAPDISK	PDSCANR	SCOMPARE	VOLFUNC
ARCHDATA	BLKPRT	EDYNAL	MAPMOD0	PDSTATS	SETRC	VSAMIO
ARCHEXTN	CLEANUP	EMACROS	MAPMOD5	PDSUNLDC	SLIM	VSAMIO1
BANNER	COBXREF	FILEDUMP	MAP3270	PDSUNLDP	SLIMAUTH	VT0C
BASALO	DATECNV	FSE	MAP3270M	PLIEXTEN	SLIMPROF	
BASCORE	DYNALO	IEBUPDTX	MYGENER	PLITERM	SLIMTSO	
BASEXTEN	DYNALOAD	JOBABEND	M327DEMO	PL1UC	TEST	
BASICMON	DYNSUB	MAKESTMT	PDSACES	PROCLIB	UTILITY	

CLEAR TO EXIT PFK03 TO RETURN

When selected, this panel shows all projects defined in the Archive. “Point and Click” is implemented here. For sake of demonstration, the cursor was placed on “SLIMTSO” and the ENTER key pressed.

SLIM Release 2 Version 1.0
February 17, 2026

----- SLIMTSO DEPENDENCIES -----
Option ==> _ 10 PROJECTS FOUND

CONNECTED TO SLIM ARCHIVE
HERCEL.SLIM.ARCHIVE

THESE ARE SUBPROJECTS FOR SLIMTSO
PRESS PFK08 TO ADD A PROJECT TO THE LIST

DYNALO	DYNSUB	EMACROS	PLIEXTEN	SLIMPROF
DYNALOAD	DYNWORK	MAP3270	SLIMAUTH	VSAMIO

CLEAR=EXIT PFK3=RETURN NOSAVE PFK12=SAVE AND RETURN PFK08 TO ADD TO LIST

This panel is a list of all project that SLIMTSO is dependent on. If something needs to be added, press PFK8 to list all eligible projects.

SLIM Release 2 Version 1.0
February 17, 2026

----- DEPENDENCIES -----
Option ==> _ 51 PROJECTS FOUND

CONNECTED TO SLIM ARCHIVE
HERCEL.SLIM.ARCHIVE

SELECT A PROJECT TO ADD TO SLIMTSO
PLACE CURSOR ON PROJECT NAME AND PRESS ENTER

\$DUMPR	BASIC1UP	DYNWORK	MAPDISK	PDSCANR	SCOMPARE	VSAMIO
ARCHDATA	BLKPRT	EDYNAL	MAPMOD0	PDSTATS	SETRC	VSAMIO1
ARCHEXTN	CLEANUP	EMACROS	MAPMOD5	PDSUNLDC	SLIM	VTOC
BANNER	COBXREF	FILEDUMP	MAP3270	PDSUNLDP	SLIMAUTH	
BASALO	DATECNV	FSE	MAP3270M	PLIEXTEN	SLIMPROF	
BASORE	DYNALO	IEBUPDTX	MYGENER	PLITERM	TEST	
BASEXTEN	DYNALOAD	JOBABEND	M327DEMO	PL1UC	UTILITY	
BASICMON	DYNSUB	MAKESTMT	PDSACES	PROCLIB	VOLFUNC	

CLEAR=EXIT PFK3=RETURN

Place the cursor on the project you wish to add and press enter. The first panel (see page 25) will reappear with the project selected here added to it.

SLIM Release 2 Version 1.0
February 17, 2026

Common Panel

The COMMON process functions very similar to the Dependencies panel. The biggest difference is you are trying to select a project to mark as COMMON. Please see Dependencies Panel (page 25) for more information.

```
----- SLIMTSO COMMON ----- USER ID = HERCEL
Option ==> _                    57 PROJECTS FOUND
```

```
Connected To Slim Archive
HERCEL.SLIM.ARCHIVE
```

```
Select A Project From The List Below To Make COMMON
Place Cursor On Project Name And Press ENTER
```

\$DUMPR	BASIC1UP	DYNCONC	JOB CARD	PDSACES	SETRC	VOLFUNC
ARCHCOMP	BASIC370	DYNINFO	MAKESTMT	PDSCANR	SIM20	VSAMIO1
ARCHDATA	BLKPRT	DYNSUB	MAPDISK	PDSTATS	SLIM	VTOT
ARCHEXTN	CATVER	DYNWORK	MAPMOD0	PDSUNLD	SLIMAUTH	
BANNER	CLEANUP	EDYNAL	MAPMOD5	PLIEXTEN	SLIMPROF	
BASALO	COBXREF	FILEDUMP	MAP3270	PLITERM	SLIMTSO	
BASCOE	DATECNV	FSE	MAP3270M	PL1UC	SYSCANR	
BASEXTEN	DYNALO	IEBUPDTX	MYGENER	PROCLIB	TEST	
BASICMON	DYNALOAD	JOBABEND	M327DEMO	SCOMPARE	UTILITY	

CLEAR TO EXIT

PFK03 TO RETURN

Batch Menu

```
----- SLIMTSO BATCH MENU -----  
Option ==> _  
  
                Connected To SLIM Archive  
                HERCEL.SLIM.ARCHIVE  
  
1  COMPILE MAP3270 Panel  
2  COMPILE and CHECK In MAP3270 Panel  
3  COMPILE a source Program  
4  COMPILE and CHECK In a source Program  
5  STORE    Non-code Items  
6  EXTRACT A Complete Project Along With Common Items*  
7  EXTRACT A Project With out Common Items*  
X  EXIT     To Previous Panel
```

* Select 6 Or 7 To Extract A Common project.

CLEAR TO QUIT PFK3 TO RETURN WITHOUT SUBMIT

This panel provides for submitting batch SLIM function. Options 1 to 5 present a panel for generating and submitting JCL to preform the requested function. Option 6 provides for project extraction.

Batch Panels

----- SLIMTSO COMPILE AND STORE PROGRAM -----

Option ==> _

OPTIONS FOR USERID HERCEL ARCHIVE HERCEL.SLIM.ARCHIVE

ADD OR UPDATE (A/U)

Group SLIMTSO

Item/Panel Name

Sub Group

Type

```
JCL To Submit
//SLIMTSO JOB CLASS=A,MSGCLASS=A,MSGLEVEL=(1,1),
//      USER=HERCEL,PASSWORD=CUL8TR,TYPRUN=SCAN
//*
//*
//*
// EXEC SLIMCLS,
// HLQ=HERCEL,USERID=HERCEL,
// CMD=,PROJECT=SLIMTSO,MODULE= ,
// DESC= ,LANG= ,
// LOADMOD= ,NCAL=,
// LOADLIB='HERCEL.SLIMTSO.NCALIB',
// IN='HERCEL.SLIMTSO.      '
//SLIMC.NOTES DD *
HERCEL 250220
//*
//*
//*
//*
//*
/*
```

CLEAR TO QUIT PFK3 TO RETURN WITHOUT SUBMIT

SLIMTSO selects which items are required for what you want to submit. The lower part of the panel displays the generated JCL. The first 5 lines are a copy of the JOB card stored in your SLIM profile. If

the job name is &JOBNAME, the project name will be substituted for &JOBNAME. You can modify anything in the first 5 lines.

Lines 6 to the end are generated based upon your entry on the menu panel. The JCL is generated every time a AID¹ key is pressed. You can change the JCL but it will be generated again and your changes lost.

Any JCL comment lines after the generated JCL can be changed and will be kept on the display as long as the number of lines in the generated jcl does not change.

Add or Update is used to tell SLIM if you are adding or updating the item. It is not used for Option 1.

Group defaults to the default project given in the SLIM Profile.

After a module name is entered and the Description is blank, a “point and click” list of valid descriptions is displayed along right side of the screen.

After a module name is entered, the Description is not blank and Type is blank, a “point and click” list of valid types for the description displayed along right side of the screen.

When every item to run the batch job, PFK12 is activated and the job can be submitted.

1 Some AID keys are ENTER, CLEAR, PF Keys, PA keys

Creating a SLIM Group

As stated back in SLIM Basics (page 5), a SLIM project consists of one or more components and sub components. A component is one or more items that generally are used together. Each item in a component is a piece of that group or a dependency on another component, called a sub group.

The key to setting up a group is to breaking it up into sub groups. For the balance of this section, I will refer to “copy code” to mean any items that are copied into COBOL programs using the COPY statement, Assembler COPY statements, user macros and PL/I INCLUDE statements.

In order to explain how a project is defined in SLIM, I will be using as an example the SLIM project as defined in SLIM.

Initialize The Group

To create an empty project for SLIM, execute proc SLIMNUPR. This will create all the possible datasets to hold the components of the project. The proc has 3 parms:

- GROUP is the project name.
- USER is the high level qualifier for the datasets, most likely will be the TSO userid.
- VOL is the volume where the datasets should be created.

For example, to create an empty project for user HERCEL called SLIM on volume TSO001:

```
//STEP EXEC SLINNUPR,USER=HERCEL,PROJECT=SLIM,VOL=TSO001
```

You can snow populate the items in the appropriate PDS. For example, using Appendix A (page 36) as guide, a new PL/1 source program will be added as a item to the .PLI PDS. Includes for the program will go in .INCLLIB.

To add this module to the Archive, run the SLIMCLS proc (page 12) or use the SLIMTSO B.4 option (Page 29).

Creation of a project is a multi step process and need not be completed all at once. The process can be dynamic and spread over the life of the development cycle. Also, consider the possible reuse of items by other projects. Items like copy or include items, macros etc. For example, I wrote some macros that provide for standard sub routine linkages. I defined the macros as a project EMACROS. To link the macros to the current project, a dependency must be established.

The item “++DEPEND” defines the relationship between a group and a subgroup. The type currently is just a filler. This group of “++DEPEND” declares that the 8 SLIM projects have a dependency on the EMACROS project.

SLIM Release 2 Version 1.0
February 17, 2026

The best way to explain the relationships is via example. This is a portion of my actual SLIM Archive.

Using the table below, the group VSAMIO has several items as part of it. VSAMIO uses the macros that are contained in group EMACROS. Therefore, a dependency exists that group VSAMIO is dependent on items from EMACROS. When an extract is requested for VSAMIO, all of its items are copied as well as the items in EMACROS.

Again, using the table below, the group SLIMTSO has a number of items (not listed). SLIMTSO uses VSAMIO so it is dependent on VSAMIO. Therefore, a dependency exists that group SLIMTSO is dependent on items from VSAMIO. VSAMIO uses the macros that are contained in group EMACROS. Therefore, a dependency exists that group VSAMIO is dependent on items from EMACROS. When an extract is requested for group SLIMTSO, all of its items are copied as well as the items in VSAMIO and EMACROS.

```

Items of Archive: HERCEL.SLIM.ARCHIVE-----
Cmd =>
C Item      Ver Group      Subgroup Type      DS RFM REC BLK      Records Date/Msg
. ++DEPEND  1 ARCHEXTN EMACROS  V00001 PS F 080 00080 1 2024-06-05
. ++DEPEND  1 BANNER EMACROS  V00001 PS FB 080 03120 1 2023-07-17
. ++DEPEND  1 BASALO EMACROS  V00001 PS FB 080 03120 1 2023-07-17
. ++DEPEND  1*FILEDUMP EMACROS V00001 PS F 080 00080 1 2024-09-29
. ++DEPEND  1 MAPDISK EMACROS  V00001 PS FB 080 03120 1 2023-08-09
. ++DEPEND  1 MAP3270 EMACROS  V00001 PS FB 080 00080 1 2023-07-17
. ++DEPEND  1 PDSACES EMACROS  V00001 PS FB 080 03120 1 2023-07-17
. ++DEPEND  1 SLIMTSO EMACROS  V00001 PS FB 080 06320 1 2025-01-10
. ++DEPEND  1 SLIMTSO PLIEXTEN V00001 PS FB 080 06320 1 2025-01-11
. ++DEPEND  1 SLIMTSO SLIMAUTH V00001 PS FB 080 06320 1 2025-01-08
. ++DEPEND  1 SLIMTSO SLIMPROF V00001 PS FB 080 06320 1 2025-01-08
. ++DEPEND  1 SLIMTSO VSAMIO  V00001 PS FB 080 06320 1 2024-11-24
. ++DEPEND  1 VSAMIO EMACROS  V00001 PS FB 080 06320 1 2024-11-24
. SLIMTSO  1 SLIMTSO SOURCE  PLI PS FB 080 06320 1 2025-03-14
. VSAMIO  1 VSAMIO COPYLIB COB PO FB 080 03120 70 2023-07-17
. VSAMIO  1 VSAMIO INCLLIB PLI PO FB 080 03120 73 2023-07-17
. VSAMIO  2 VSAMIO LOADLIB NCAL PO U 000 19069 8 2024-03-06
. VSAMIO  1 VSAMIO LOADLIB NCAL PO U 000 19040 8 2024-01-03
. VSAMIO  1 VSAMIO MACRO ASM PS FB 080 03120 116 2024-03-06
. VSAMIO  2 VSAMIO SUBPGM ASM PS FB 080 03120 727 2024-03-06
. VSAMIO  1 VSAMIO SUBPGM ASM PS FB 080 03120 671 2024-01-03

```

When an SLIM extract process (SLIMEXTR Proc on page 15) for a group is executed, the Archive is scanned and a list of all the items the a group is created. All items in the subgroups specified in the “++DEPEND” are included in this list. The list is then used to produce JCL to extract all the items from the archive and JCL to compile the entire group is generated. It is possible that sub-groups can have dependencies. These will be processed as well.

The extract process always select the most current version of an item. It is possible to have conflicts between items in different groups. This occurs when an item exists in two or more groups. Extract process will detect that, issue a message and stop the extract process. These conflicts must manually be resolved.

First, the components must be added to the archive. This can be done using the SLIMxxxx procs (see Using SLIM on page 10). Next the dependencies must be defined using the SLIMDEP proc (see page 15).

First, look through all the code for any copy code. These would be candidates for sub groupss. For example, I have some user macros I wrote and use in almost all the assembler programs. These macros would be candidates for a sub group.

ITEMS That Require Special Handling

The CLEANUP assembler program requires macros that are not in SYS1.MACLIB, an additional PDS SYS1.AMODGEN is included in the SLIM procs for a clean assembly.

Any PL/I (F) program such as SLIMXREF that uses the internal sort feature will have a link edit return code of 4. Normally the return code (rc) other than 0 indicates a problem with the link edit and proc SLIMCLS will abend. This is caused by the way a PL/I runtime module was link edited. However, the program will still execute correctly. A SLIM type of 'PLIS' is included to handle this situation. With an item with type PLIS is compiled, SLIMCLS will not abend with this condition.

Customizing SLIM

The full source code for SLIM and all modules it depends on are supplied. SLIM is designed for easy customization. In order to customize, you must understand how SLIM works.

Most important to note is that these tables are coded as assembler programs. Any changes requires that the table be re-assembled and link edited. For TK5, the SYSLIB DD must include the SLIM.MACLIB. For other versions of MVS, the SYSLIB DD must include the PDS containing the SLIMDDN and SLIMAUTH macros.

To avoid re-compiling SLIM and SLIMTSO whenever these tables change, they are implemented as executable modules. They are loaded into memory each time SLIM or SLIMTSO are executed.

Each item stored has a module (item) name, a group name, a description and a type. Item names and group names are not validated. The description and type are validated using a table.

SLIMDDN Table

Each description tells about what the item is and type specifies the nature of the item (i.e assembler source program). The DDN indicates which dataset will get the item during an extract process.

Referring to appendix A on page 36, during the extract process, a JCL stream is generated to write all the items associated with a group to an appropriate PDS. For example, COPYLIB for type COB will be written to JCL DDNAME COPYCOB.

The generated JCL utilizes proc SLIMEXT2. Proc SLIMEXT2 has a DD statement for each DDNAME listed in the table. If any DDN is changed or added to the table, the corresponding DD statement must be changed or added to the proc SLIMEXT2.

SLIMAUTH Table

SLIM has the ability to dynamical allocate ARCHIVER files, it has a builtin “security” system. All executions of SLIM require a HLQ (high level qualifier and a USER KEY. HLQ is usually a TSOID or catalog alias. USER KEY is a unique name assigned to each user authorized to access SLIM archives. It can be any combination of up to 8 characters. There is a module called SLIMAUTH that verifies the HLQ and USERID to determine the dataset name of the ARCHIVE to be accessed.

Appendix B on page 37 is a sample of the SLIMAUTH module contained in the distribution.

Appendix A – The SLIMDDN Table

This is the table as delivered with SLIM.

```

      TITLE '*** SLIMDDN V2.0.0 ***'
*****
*
*   SLIMDDN - SLIM DDNAME TABLE
*
*   **** THIS MODULE IS USED BY SLIM AND SLIMTSO PROGRAMS. ****
*
*   THIS MODULE ACCEPTS THE SLIM DESCRIPTION AND TYPE TO DETERMINE
*   THE IF THE COMBINATION IS VALID AND IF SO THE DDNAME FOR JCL.
*
*   =====> NOTICE <=====
*   THERE IS A GROUP OF SPECIAL DESC/TYPE COMBINATION THAT ARE
*   REQUIRED BY SLIM.  THESE ARE GENERATED BY DEFAULT.  IF ANY
*   OF THESE ARE CHANGED, THE SLIM MODULE MAY NEED TO BE MODIFIED.
*   THESE CANNOT BE OVER RIDEN.
*
*   THE GROUP IS:
*       COPYLIB  COB    COPYCOB
*       COPYLIB  ASM    COPYASM
*       INCLLIB  PLI    INCLLIB
*       INCLLIB  PLIS   INCLLIB
*       MACRO    ASM    MACLIB
*       SOURCE   PLI    PLI
*       SOURCE   PLIS   PLI
*       SUBPGM   PLI    PLI
*       SUBPGM   PLIS   PLI
*
*****
*
*   SLIMDDN TYPE=START
*
*   SLIMDDN  DESC=INSTALL,DTYPE=CNTRL,DDN=JCL
*   SLIMDDN  DESC=LOADLIB,DTYPE=EXEC,DDN=LOADLIB
*   SLIMDDN  DESC=LOADLIB,DTYPE=NCAL,DDN=NCALIB
*   SLIMDDN  DESC=PROCLIB,DTYPE=JCL,DDN=PROCLIB
*
*   SLIMDDN  DESC=SOURCE,DTYPE=ASM,DDN=ASM
*   SLIMDDN  DESC=SOURCE,DTYPE=COB,DDN=COB
*   SLIMDDN  DESC=SOURCE,DTYPE=FOR,DDN=FOR
*   SLIMDDN  DESC=SOURCE,DTYPE=PAS,DDN=PAS
*
*   SLIMDDN  DESC=MAP,DTYPE=ASM,DDN=MAP
*   SLIMDDN  DESC=PANEL,DTYPE=DATA,DDN=PANEL
*   SLIMDDN  DESC=PANEL5,DTYPE=DATA,DDN=PANEL5
*   SLIMDDN  DESC=SOURCE5,DTYPE=PLI,DDN=SOURCE5
*
*   SLIMDDN  DESC=SUBPGM,DTYPE=ASM,DDN=ASM
*   SLIMDDN  DESC=SUBPGM,DTYPE=COB,DDN=COB
*   SLIMDDN  DESC=SUBPGM,DTYPE=FOR,DDN=FOR
*   SLIMDDN  DESC=SUBPGM,DTYPE=PAS,DDN=PAS
*
*   SLIMDDN  DESC=SYSOUT,DTYPE=LISTING,DDN=LISTING
*   SLIMDDN  DESC=SYSOUT,DTYPE=REPORT,DDN=REPORTS
*   SLIMDDN  DESC=DOCUMENT,DTYPE=TEXT,DDN=DOC
*   SLIMDDN  DESC=TEXT,DTYPE=TEXT,DDN=TEXT
*
*   SLIMDDN TYPE=END
*   END

```

Appendix B – The SLIMAUTH Table

This is the table as delivered with SLIM.

```

      TITLE '*** SLIMAUTH V2.0.0 ***'
*****
*
*   SLIMAUTH/SLIMAUT - SLIM AUTHORIZATION MODULE
*
*   **** THIS MODULE IS USED BY SLIM AND SLIMTSO PROGRAMS. ****
*
*   THIS MODULE ACCEPTS THE SLIM USER ID (USR) AND HIGH LEVEL
*   QUALIFIER (HLQ). BOTH MUST BE DEFINED IN THIS MODULE.
*   TOGETHER THE SLIM ARCHIVE THAT IS TO BE ACCESSED CAN BE
*   DETERMINED.
*
*   FOR SLIM, IF THE COMBINATION IS NOT DEFINED, SLIM WILL
*   ABEND CODE U0001 OR U0002.
*   FOR SLIMTSO, IF THE COMBINATION IS NOT DEFINED, SLIMTSO WILL
*   ISSUE A MESSAGE AND USER PROMPTED TO ENTER AGAIN.
*
*   NOTE THE SLIM USER ID IS NOT THE SAME AS THE TSO USER ID.
*
*   THIS IS THE TABLE USED DISTRIBUTED WITH SLIM.
*   NOTE THE TYPE MUST BE IN THE SEQUENCE AS SHOWN BELOW.
*   THE HLQ AND USER MUST BE 8 CHARACTER OR LESS.
*   ARCH IS THE DATASET NAME OF THE ARCHIVE THIS COMBINATION OF
*   USER AND HLQ CAN ACCESS.
*
*****
SLIMAUTH TYPE=START
*
SLIMAUTH TYPE=HLQ,HLQ=HERC01
*
SLIMAUTH TYPE=USER,HLQ=HERC01,USER=DEFAULT,ARCH=HERC01.SLIM.ARCHIVE
SLIMAUTH TYPE=USER,HLQ=HERC01,USER=HERC01,ARCH=HERC01.SLIM.ARCHIVE
*
SLIMAUTH TYPE=END
END
```